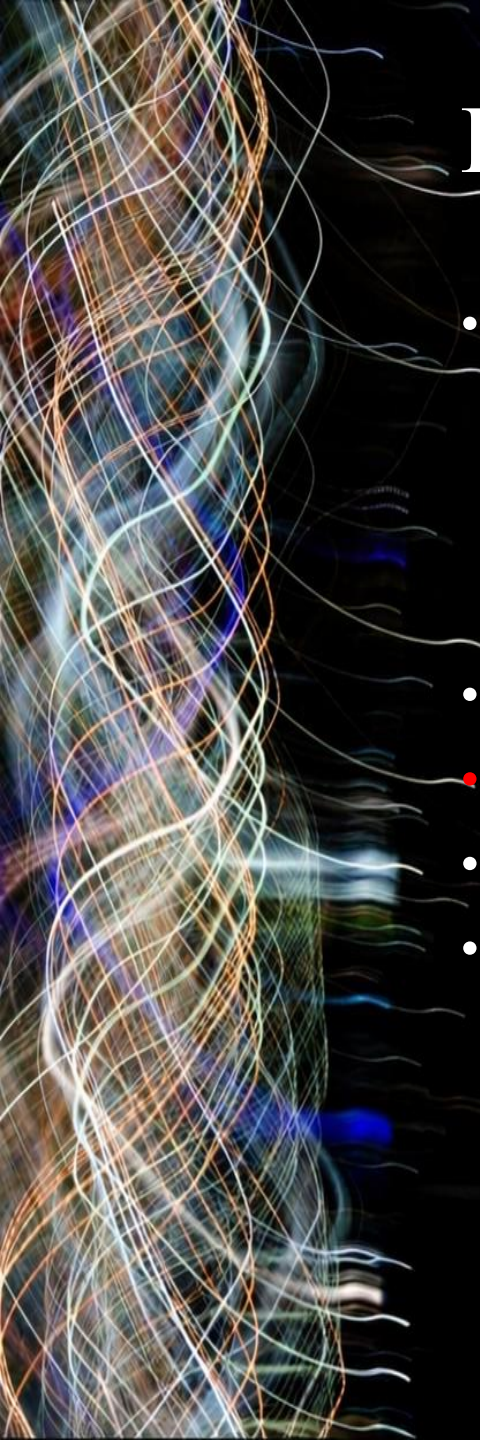


超導量子位元的特性、 超導量子電路

王培儒

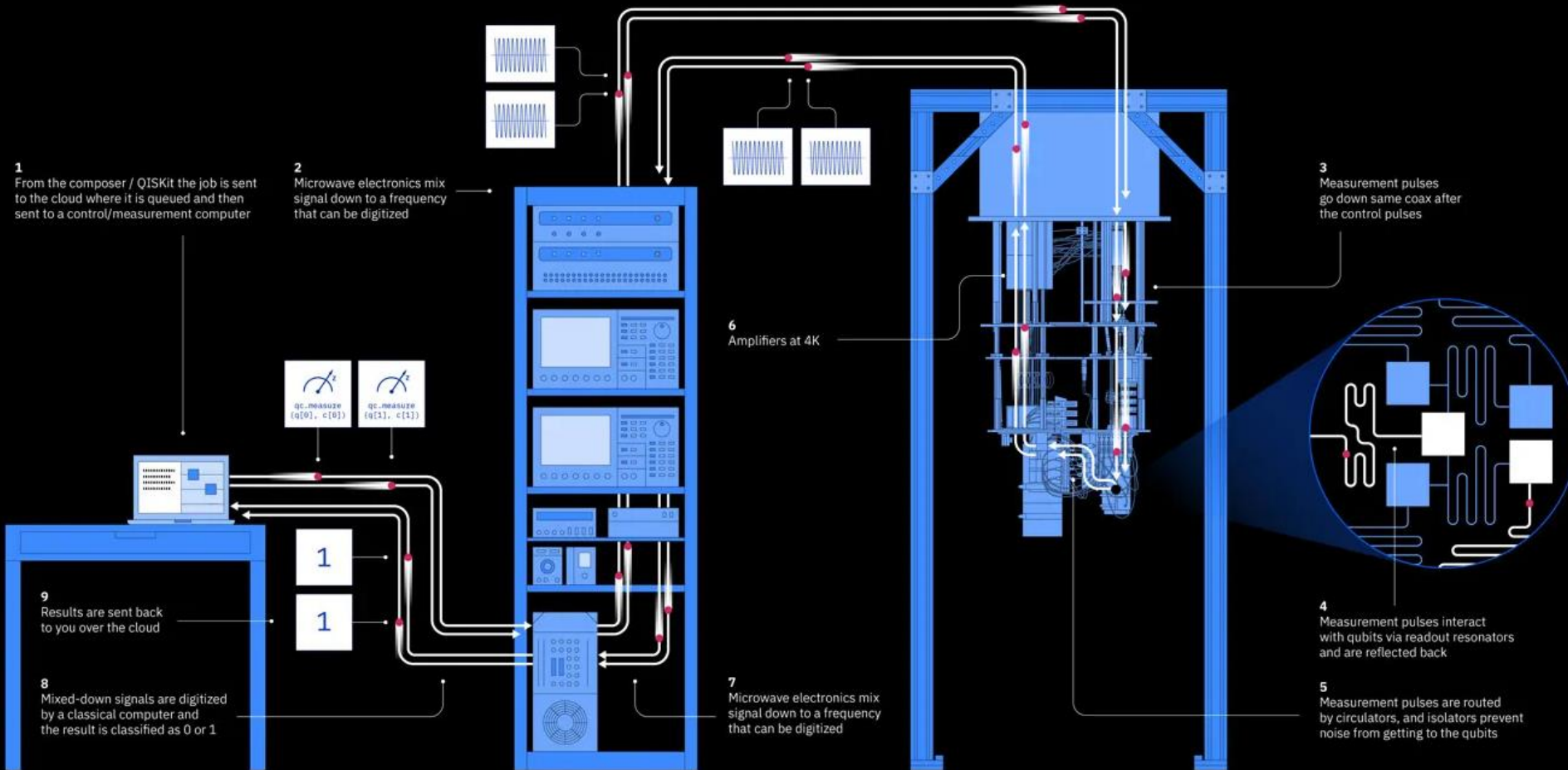
2026.04.27



Key Information

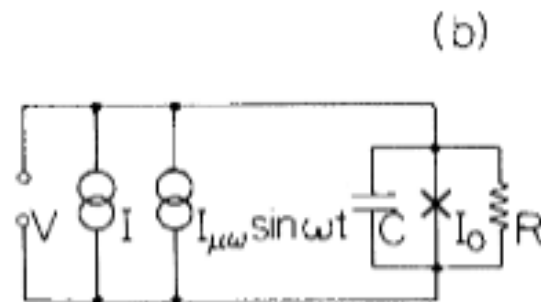
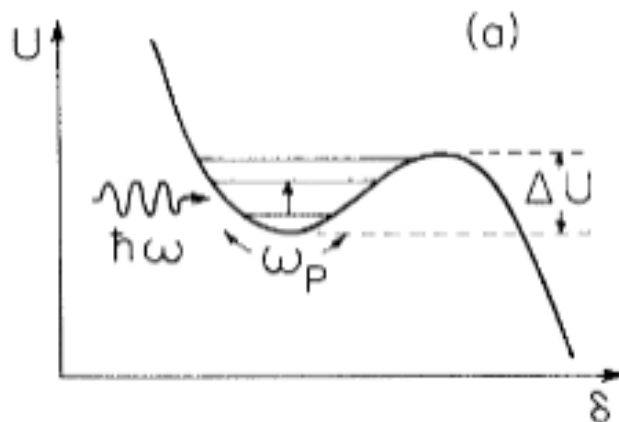
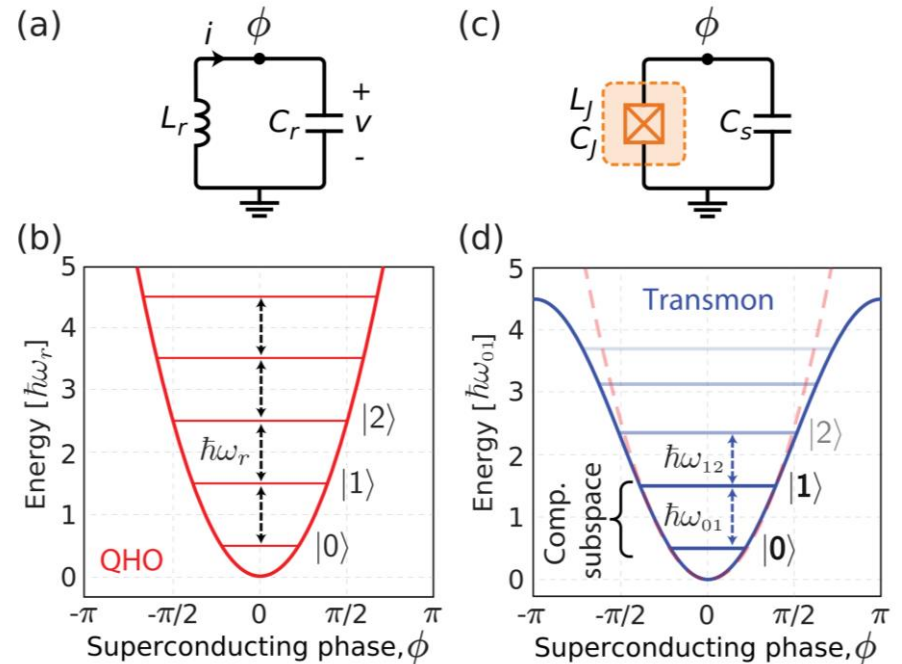
- 超導量子計算(Superconducting Quantum Computation, SC QC)，是在極低溫下，以超導(Superconductor)製作的約瑟夫森節(Josephson Junction)為基礎，設計出非線性的LC電路作為量子位元 (Qubit)，搭配電路設計做出可控制、可讀出、可計算的量子處理器(QPU)。整機運作結合微波工程、低溫物理、材料、電路設計與量子控制。
- 04/20：超導量子計算是什麼？
- 04/27：超導量子位元的特性、超導量子電路
- 05/04：如何控制、校正超導量子位元？
- 05/11：超導量子電腦硬體元件導論

IBM's Prototype

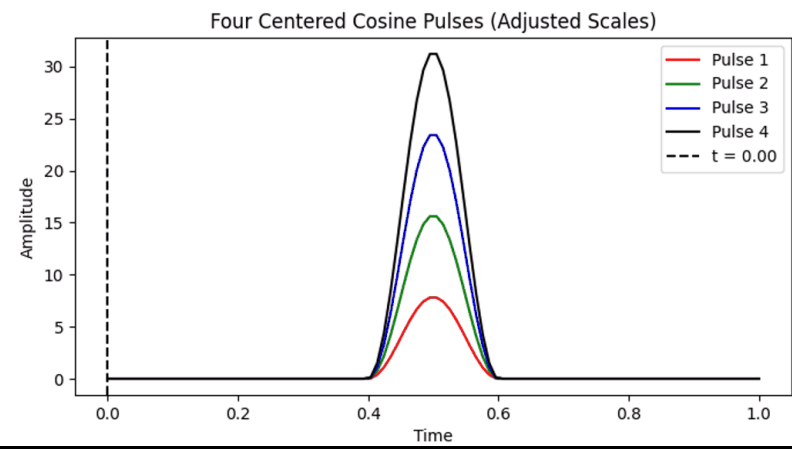
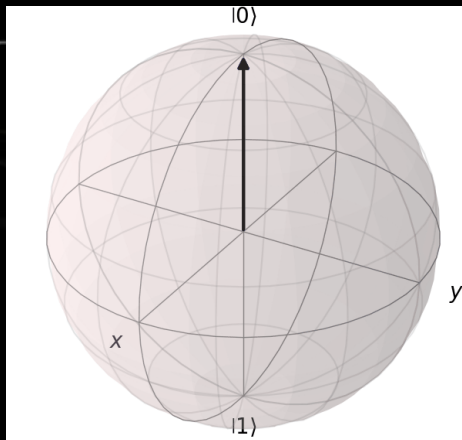
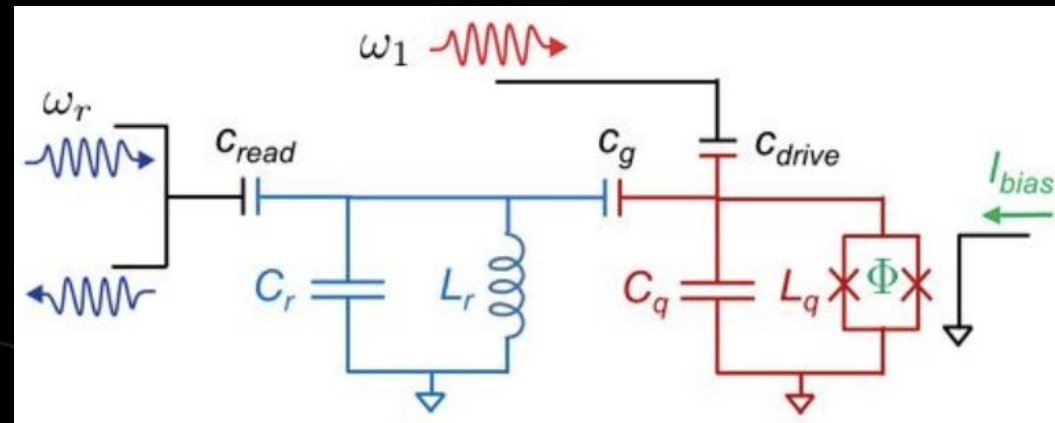
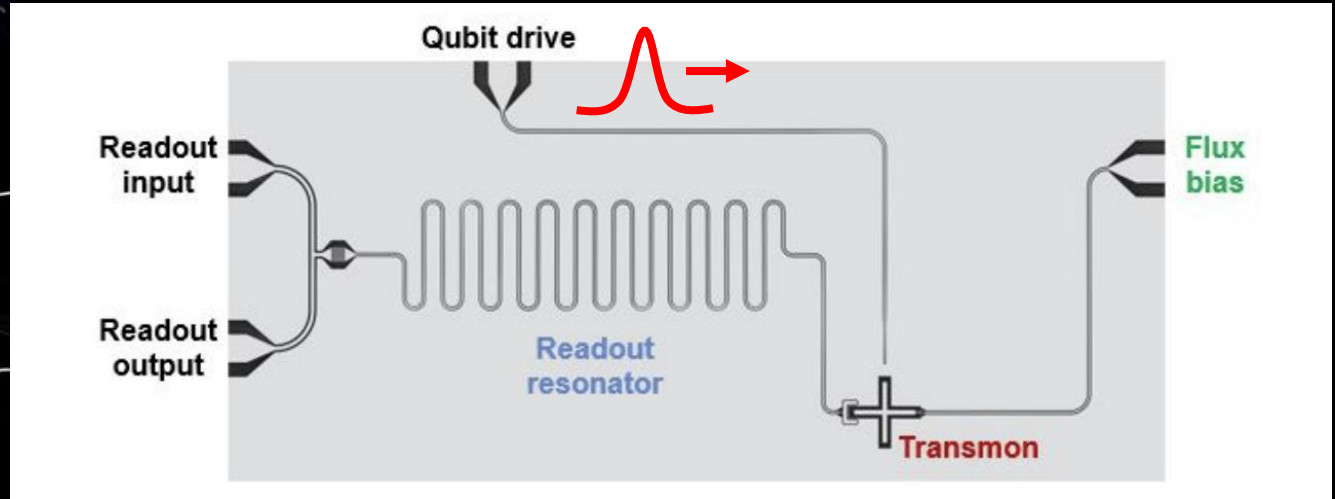


超導量子位元-非線性LC電路

10.1103/PhysRevLett.55.1543, 10.1063/1.5089550, Nobel



1985





Quantum Control

量子模擬



Simulating Physics with Computers

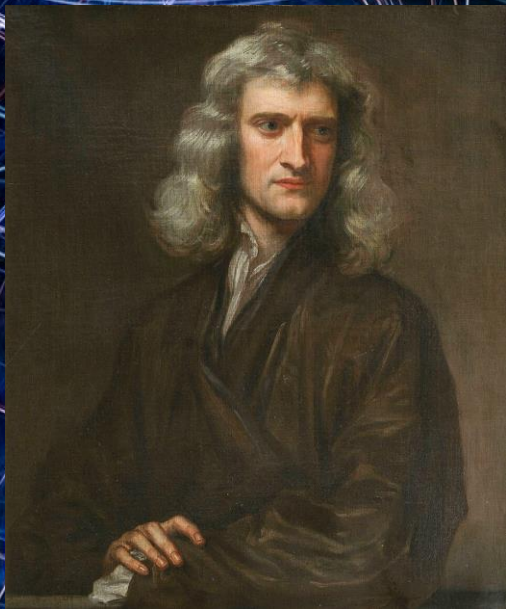
Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

be understood very well in analyzing the situation. And I'm not happy with all the analyses that go with just the classical theory, because nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy. Thank you.

物理定律(通常)是可逆的



Newton

$$F = \dot{p}$$



Hamilton

$$\dot{q} = \frac{\partial H}{\partial p}$$
$$\dot{p} = -\frac{\partial H}{\partial q}$$



Schrodinger

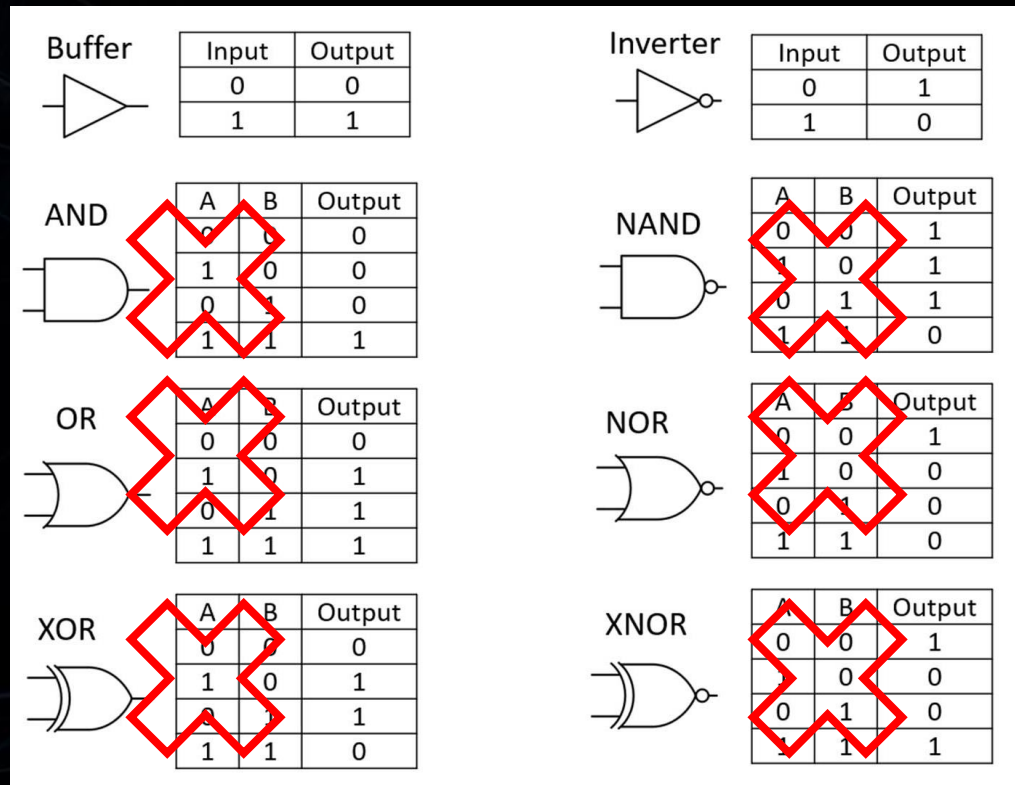
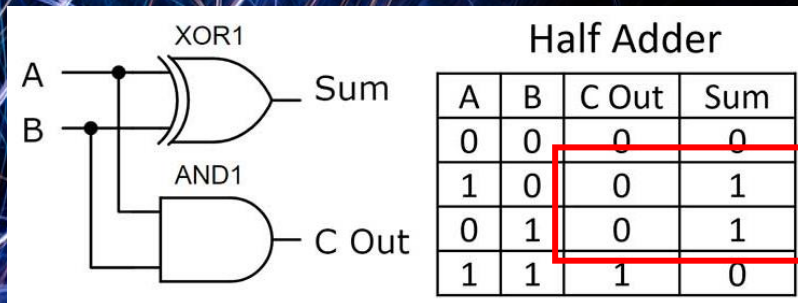
$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

$$\psi(t) = \left[\exp \left(-\frac{i}{\hbar} T \left[\int_0^t \hat{H}(s) ds \right] \right) \right] \psi(0)$$

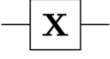
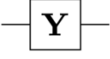
資訊抹除與不可逆性

- 抹除或喪失1 bit的資訊，會至少產生 $k_B \ln 2$ 的熵
(或需要耗能 $E \geq k_B T \ln 2$)

Landauer



可逆邏輯閘-Single Qubit Gate

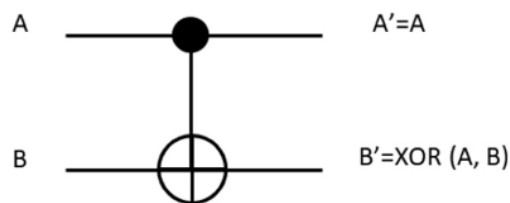
X Gate (NOT Gate)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$ 0\rangle \rightarrow 1\rangle, 1\rangle \rightarrow 0\rangle$
Y Gate			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Z Gate			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Hadamard Gate		Superposition	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$ 0\rangle \rightarrow \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$

Control-NOT Gate (CNOT)/Control-X Gate (CX)

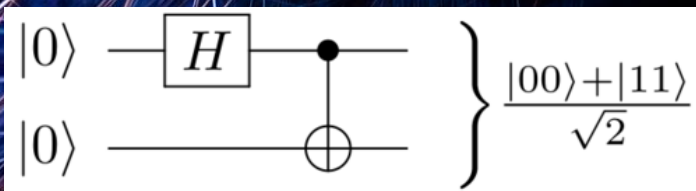
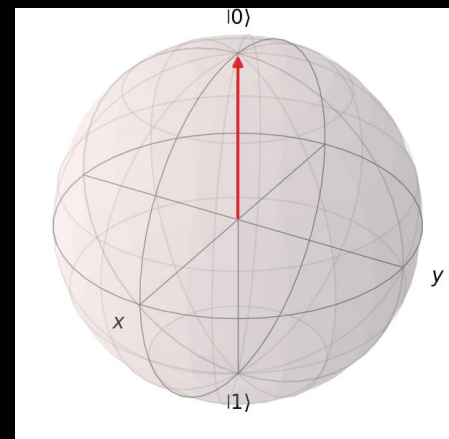
Logic Table

A	B	A'	B'
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Gate Diagram



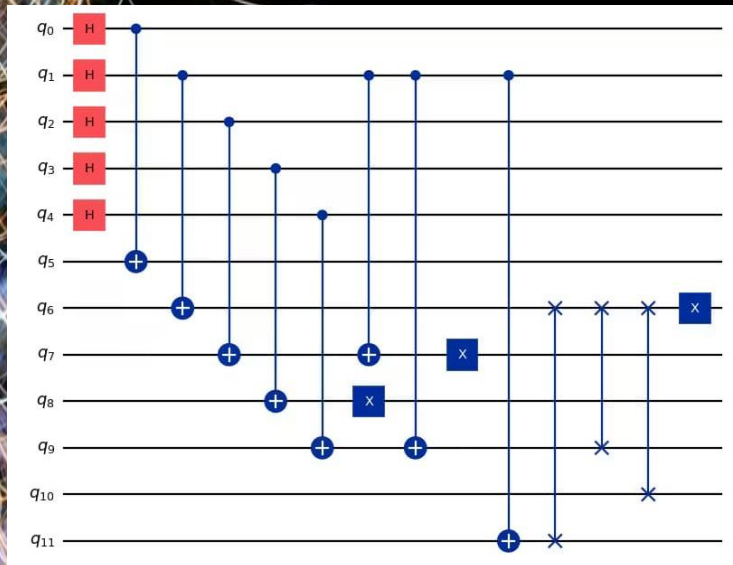
X Gate (NOT Gate)



$$|0\rangle_A |0\rangle_B \rightarrow \frac{(|0\rangle_A + |1\rangle_B)}{\sqrt{2}} |0\rangle_B \rightarrow \frac{(|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B)}{\sqrt{2}}$$

Entanglement/Bell state

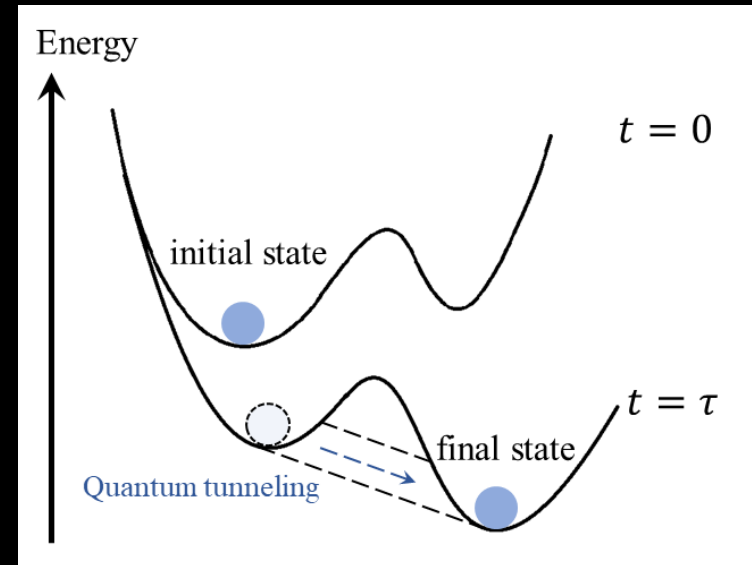
Gate Based/Circuit Model



- Algorithm: Shor, Grover,...
- Quantum Simulation
- Discrete control (logic gate)



Adiabatic/Annealing /Ising Machine



- Optimization: salesman, max-cut...
- Ground State Search
- Continuous control

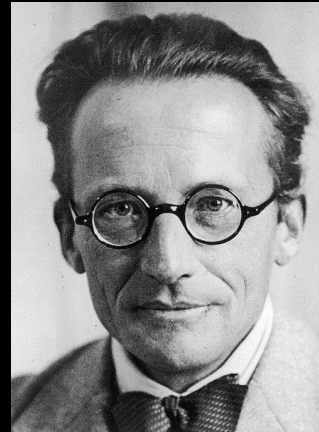
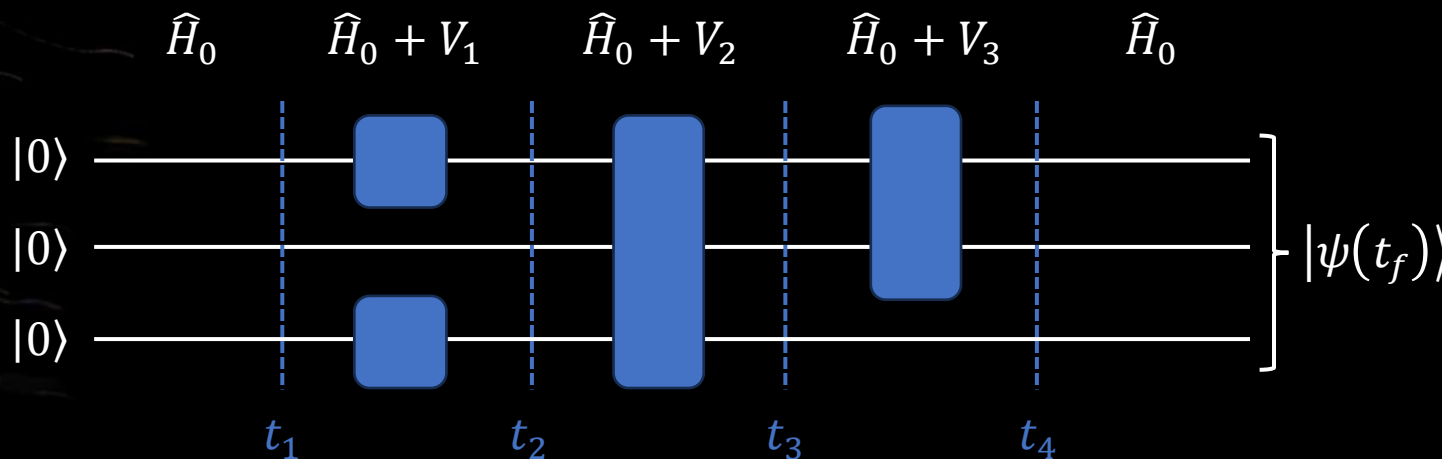


Quantum Time Evolution

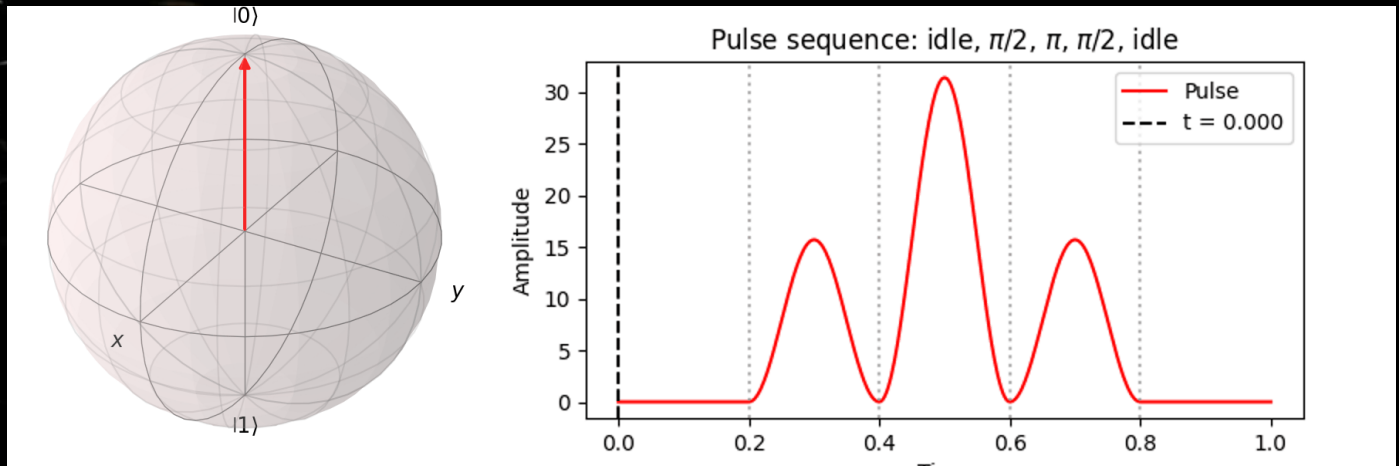
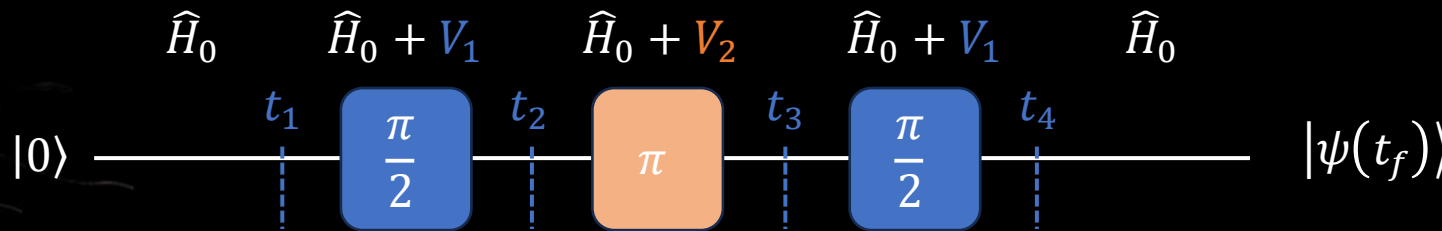
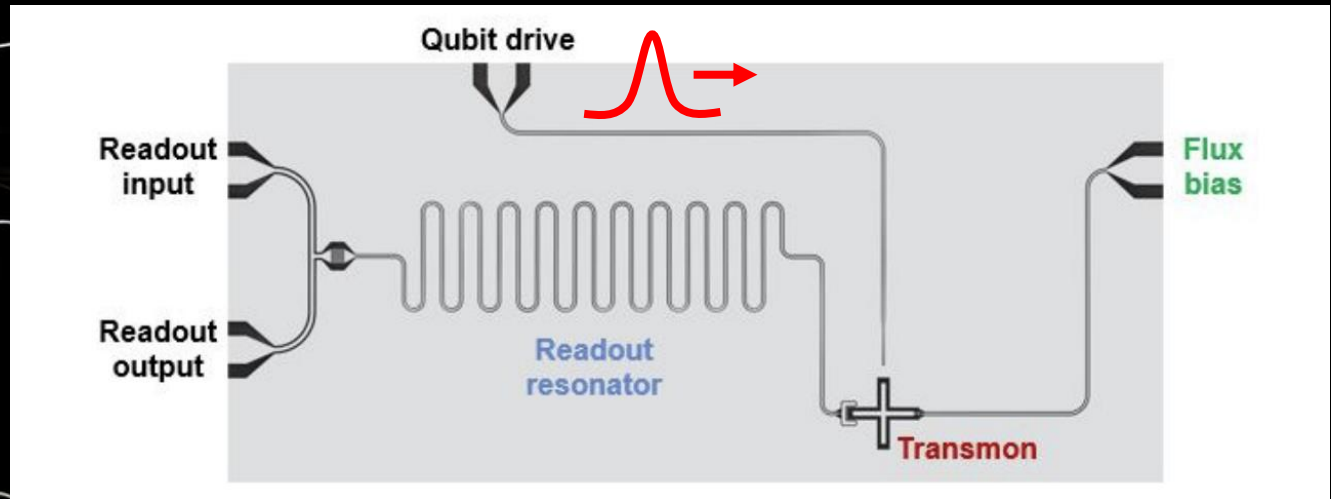
$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

$$\psi(t_f) = \underbrace{\left[\exp \left(-\frac{i}{\hbar} T \left[\int_{t_i}^{t_f} \hat{H}(s) ds \right] \right) \right]}_{U(t_f, t_i)} \psi(t_i)$$

$$= U(t_f, t_4) * U(t_4, t_3) * U(t_3, t_2) * U(t_2, t_1) * U(t_1, t_i) * \psi(t_i)$$

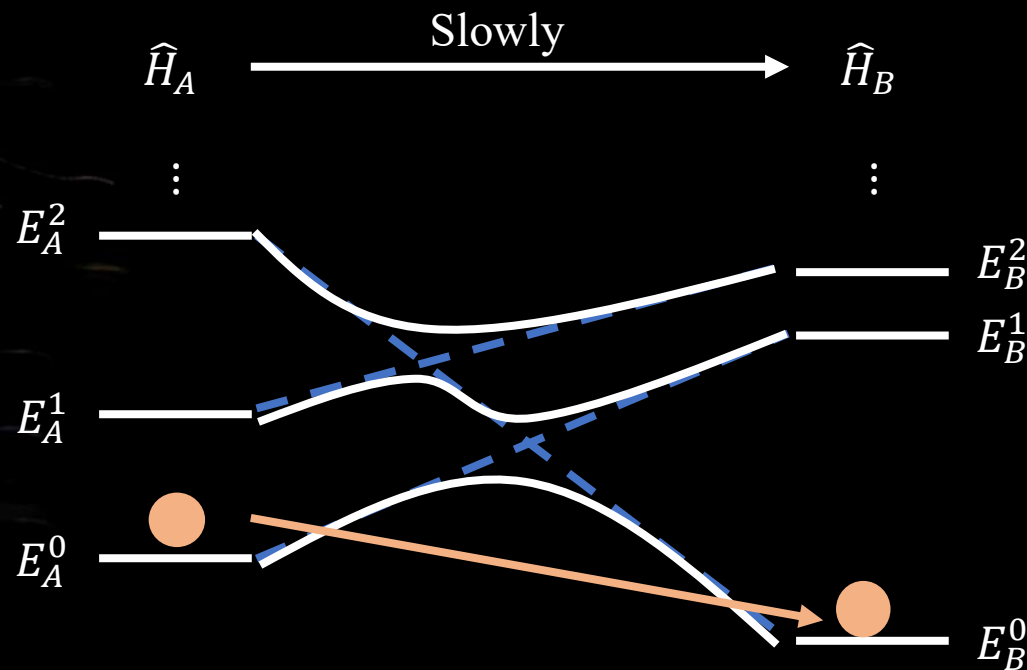
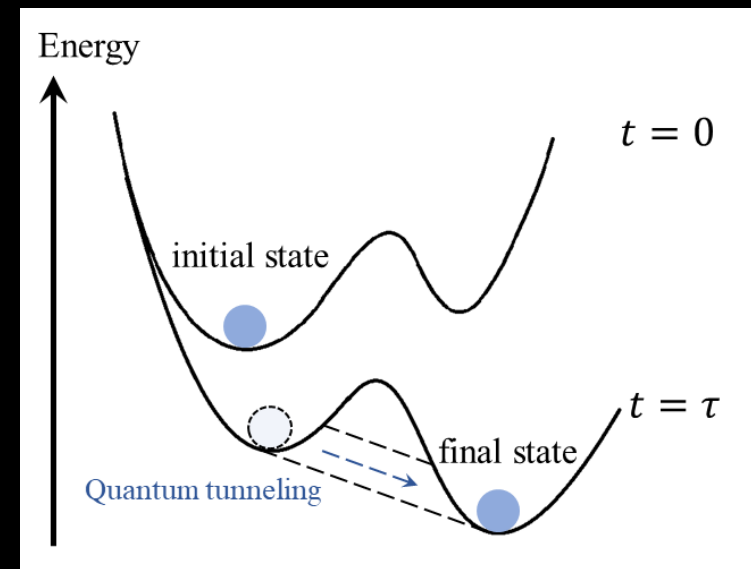


$$\psi(t_f) = U(t_f, t_4) * U(t_4, t_3) * U(t_3, t_2) * U(t_2, t_1) * U(t_1, t_i) * \psi(t_i)$$



Adiabatic QC

$$\hat{H}(t) = \begin{cases} \hat{H}_A & , t < 0 \\ \left(1 - \frac{t}{T}\right) \hat{H}_A + \frac{t}{T} \hat{H}_B & , 0 < t < T \\ \hat{H}_B & , t > T \end{cases}$$



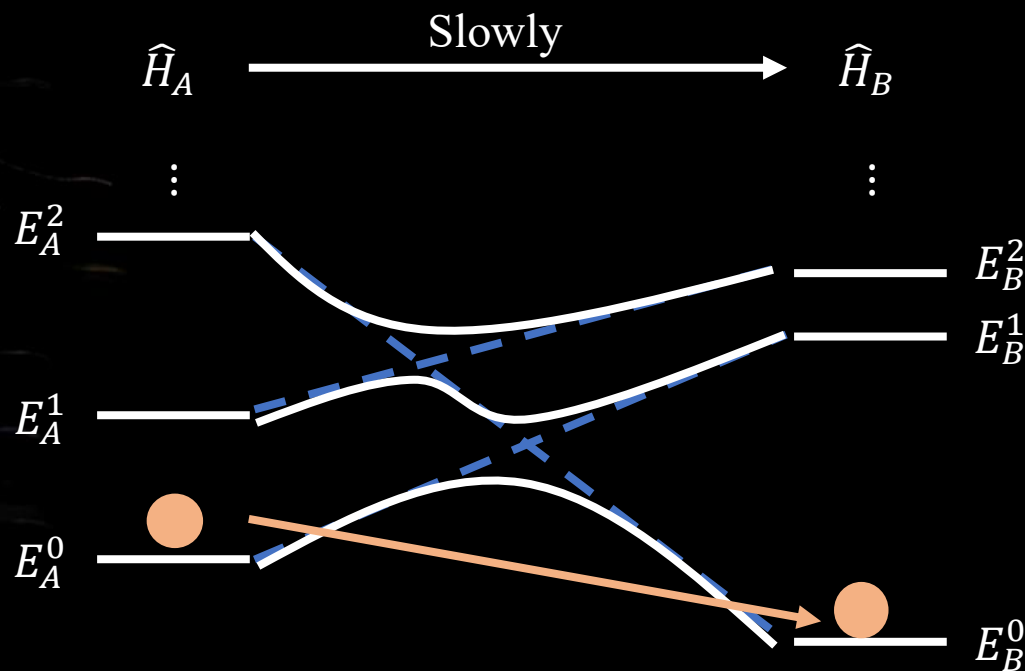
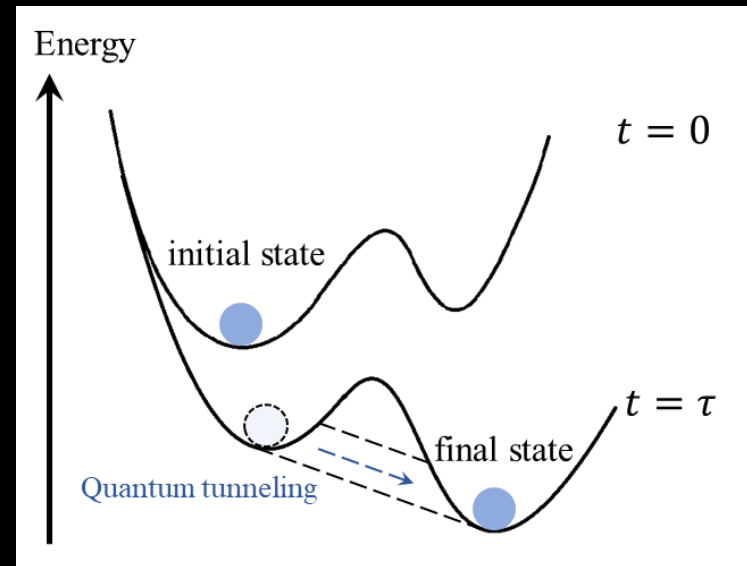
Adiabatic QC

$$0 < t < T$$

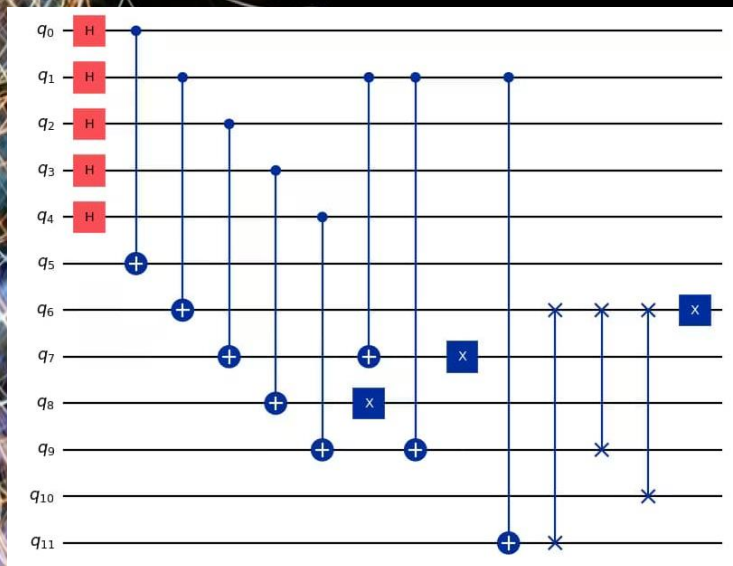
$$\hat{H}(t) = \left(1 - \frac{t}{T}\right) \hat{H}_A + \frac{t}{T} \hat{H}_B$$

$$= \hat{H}_A + \frac{t}{T} (\hat{H}_B - \hat{H}_A)$$

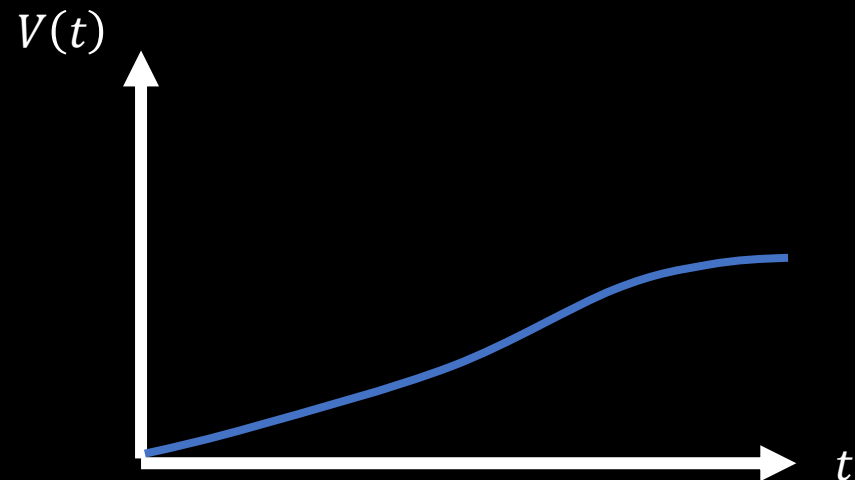
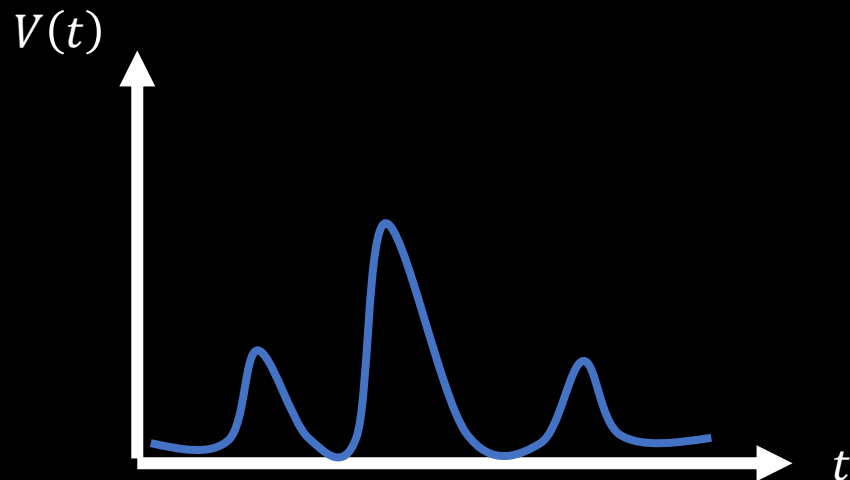
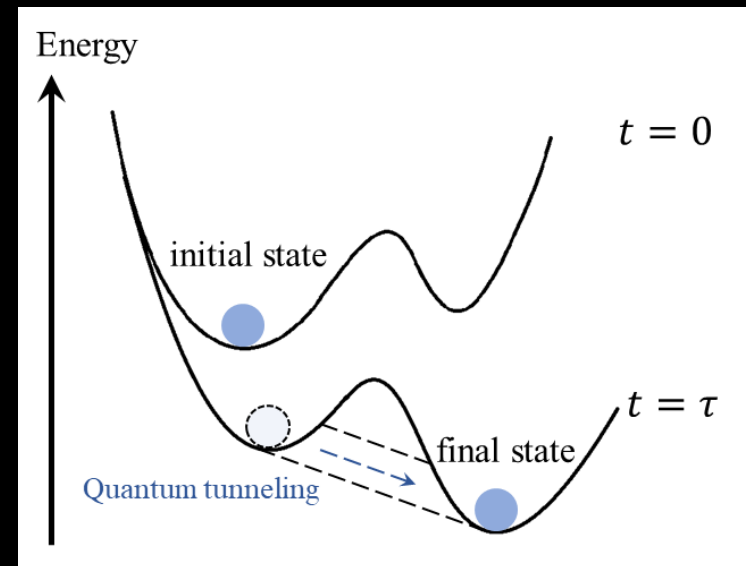
$$= \hat{H}_A + V_T(t)$$



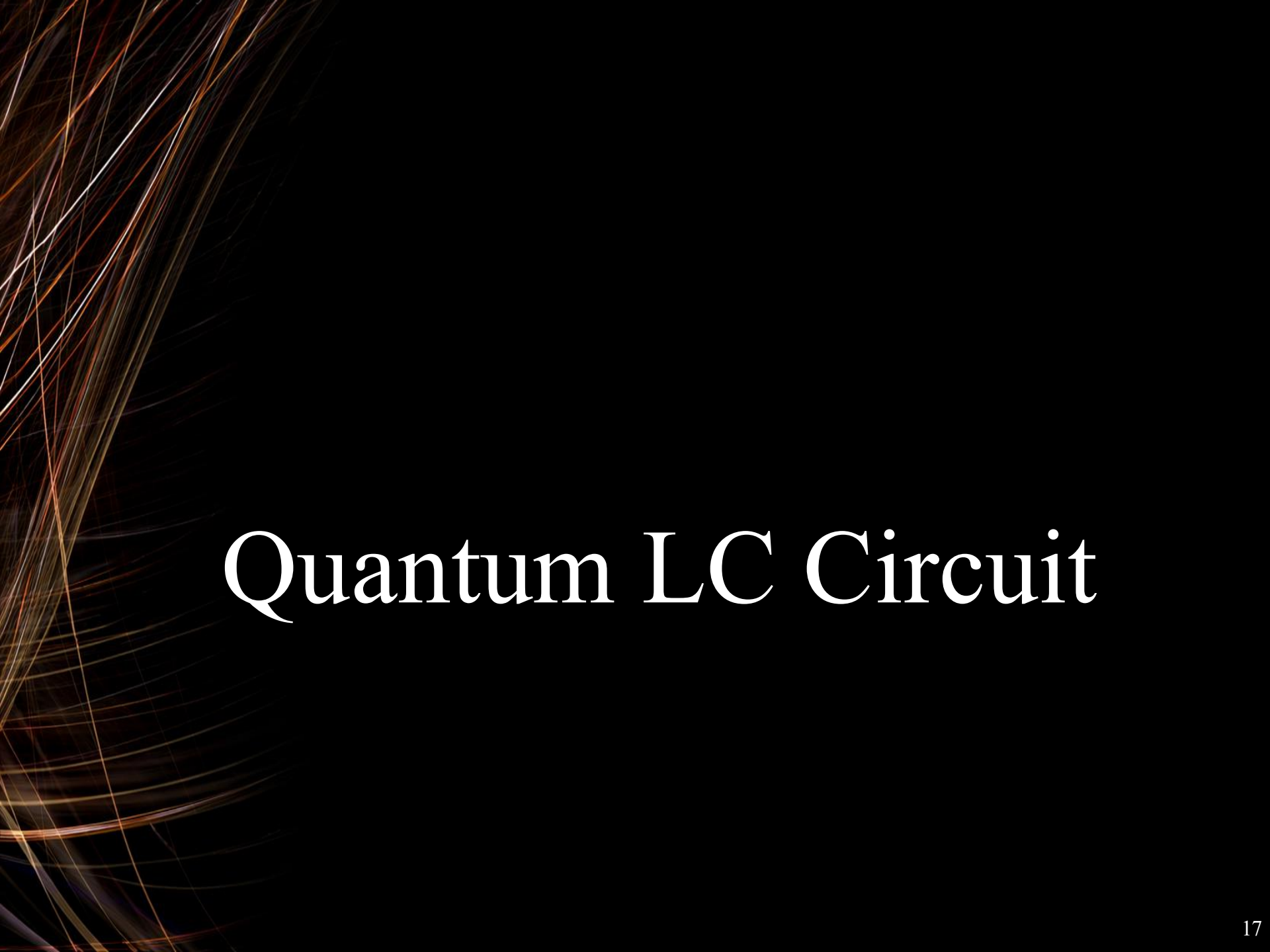
Gate Based/Circuit Model



Adiabatic/Annealing /Ising Machine

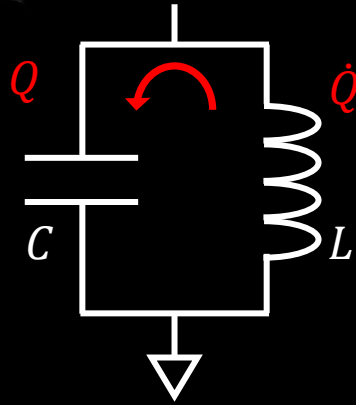


Adiabatic Quantum Computation is Equivalent to Standard Quantum Computation (2008) 16

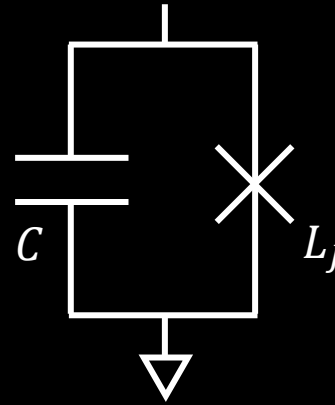


Quantum LC Circuit

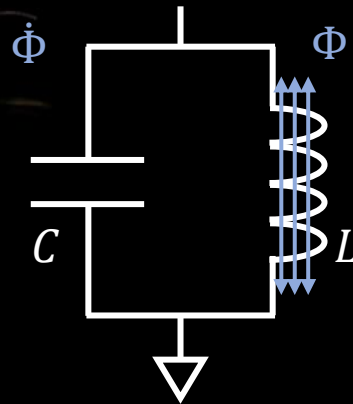
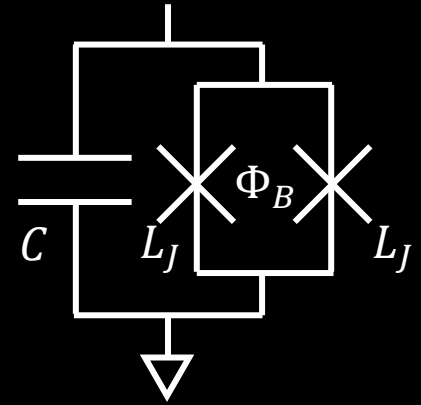
LC Circuit



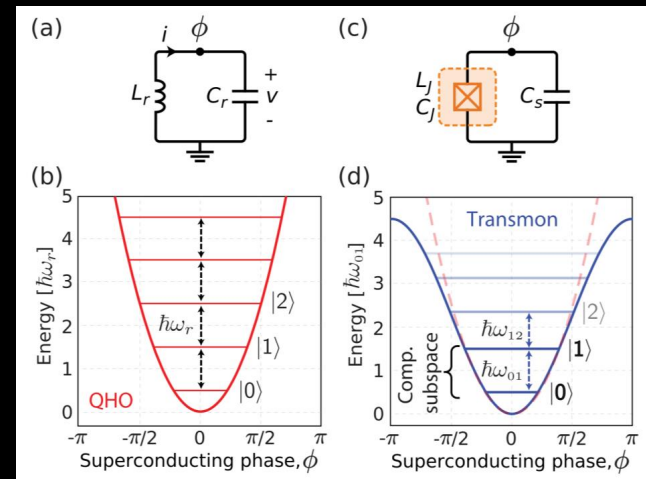
$$E = \frac{Q^2}{2C} + \frac{L\dot{Q}^2}{2}$$



*Non – linear Kinetic Energy
→ Hard to understand*



$$E = \frac{C\dot{\Phi}^2}{2} + \frac{\Phi^2}{2L}$$

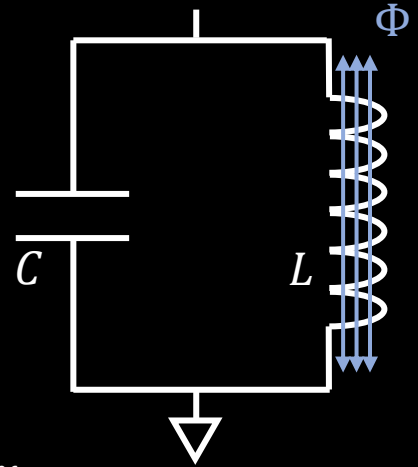


Non – linear Potential Energy

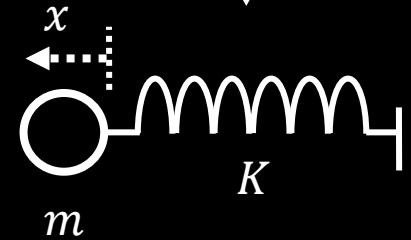
$$E = \frac{C\dot{\Phi}^2}{2} + \frac{\Phi^2}{2L(\Phi_B)} - \beta(\Phi_B)\Phi^4 + \dots$$

LC Circuit

$$\begin{cases} \text{Capacitor : } I_C = C \frac{dV_C}{dt} \\ \text{Inductor : } V_L = L \frac{dI_L}{dt} \end{cases} \rightarrow E = \int_0^t IV dt$$



$$\rightarrow \begin{cases} E_C = \frac{Q^2}{2C} \\ E_L = \frac{L\dot{Q}^2}{2} \end{cases} \leftrightarrow \begin{cases} V = \frac{1}{2}Kx^2 \\ T = \frac{1}{2}m\dot{x}^2 \end{cases}$$



$$\begin{aligned} \Phi &\equiv \int_0^t V_L dt = \int_0^t L \frac{dI_L}{dt} dt = L * \dot{Q} \\ \frac{d\Phi}{dt} &= V_L = V_C = \frac{Q}{C} \rightarrow Q = C\dot{\Phi} \end{aligned} \rightarrow \begin{cases} E_C = \frac{C\dot{\Phi}^2}{2} \\ E_L = \frac{\Phi^2}{2L} \end{cases}$$

$$\text{Lagrangian } \mathcal{L} = \frac{C\dot{\Phi}^2}{2} - \frac{\Phi^2}{2L}$$

$$\text{Conjugate Momentum } \Pi = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi} = Q$$

$$\text{Hamiltonian } \mathcal{H} = \Pi\dot{\Phi} - \mathcal{L} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

Canonical Quantization
 $[\hat{\Phi}, \hat{Q}] = i\hbar \leftrightarrow [x, p] = i\hbar$

LC Circuit

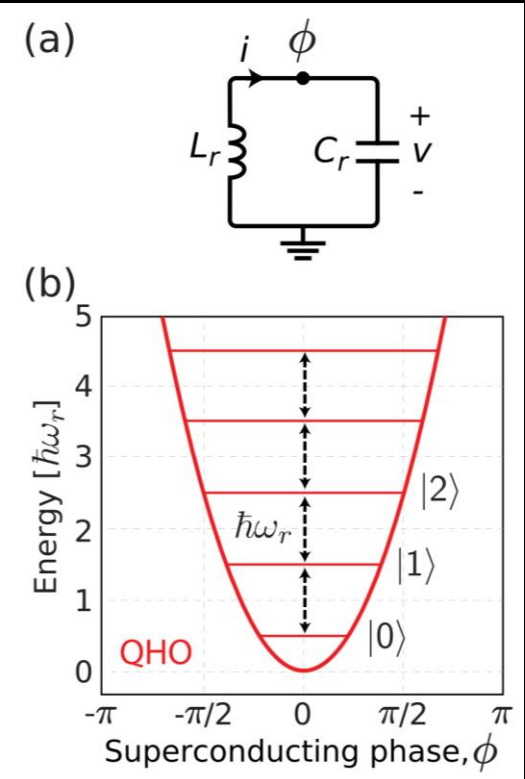
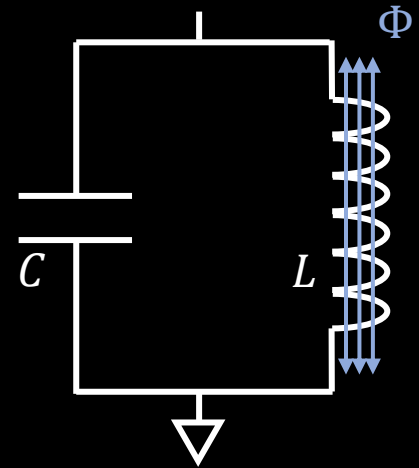
$$\text{Hamiltonian } \hat{\mathcal{H}} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}$$

Canonical Quantization
 $[\hat{\Phi}, \hat{Q}] = i\hbar$

$$\begin{cases} \hat{a} = \sqrt{\frac{1}{2\hbar}} \sqrt{\frac{L}{C}} \left(\sqrt{\frac{C}{L}} \hat{\Phi} + i\hat{Q} \right) \\ \hat{a}^\dagger = \sqrt{\frac{1}{2\hbar}} \sqrt{\frac{L}{C}} \left(\sqrt{\frac{C}{L}} \hat{\Phi} - i\hat{Q} \right) \end{cases}$$

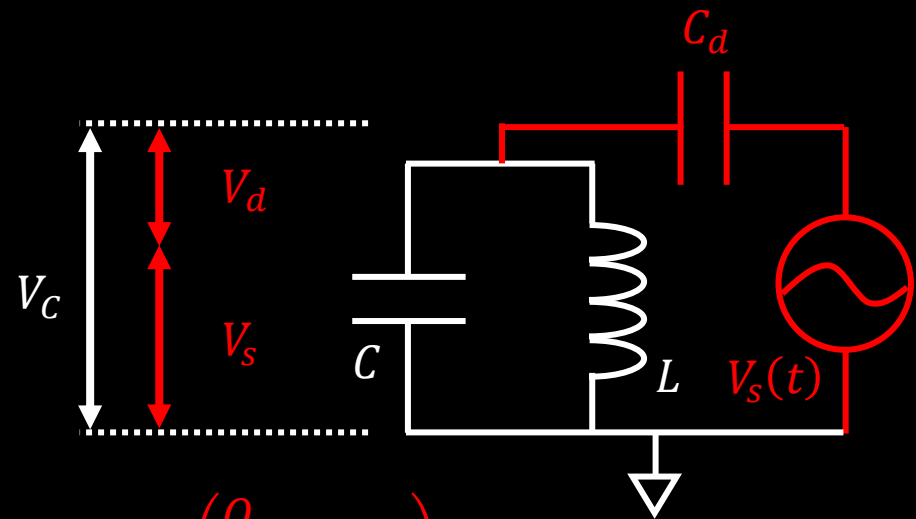
$$\hat{\mathcal{H}} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

$$\omega = \frac{1}{\sqrt{LC}}$$



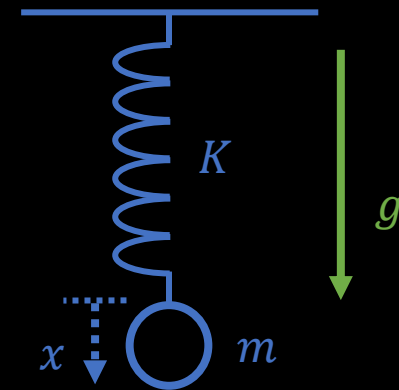
Driven

$$\mathcal{H} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} + \frac{Q_d^2}{2C_d}$$



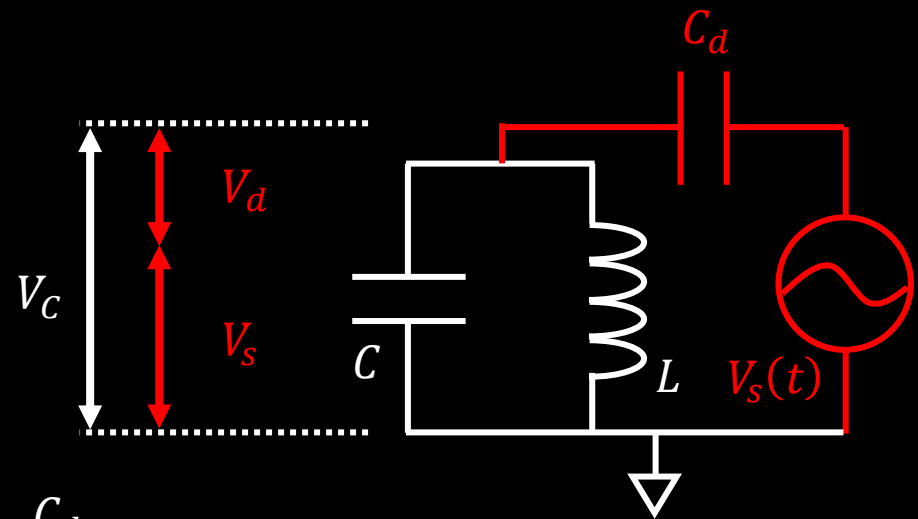
$$Q_d = C_d V_d = C_d (V_C - V_s(t)) = C_d \left(\frac{Q}{C} - V_s(t) \right)$$

$$\begin{aligned} \mathcal{H} &= \frac{Q^2}{2C} + \frac{\Phi^2}{2L} + \frac{C_d^2 \left(\frac{Q}{C} - V_s(t) \right)^2}{2C_d} \\ &= \frac{Q^2}{2C} + \frac{\Phi^2}{2L} + \frac{C_d}{2} \left(\frac{Q^2}{C^2} - 2 \frac{Q}{C} V_s(t) + V_s^2 \right) \\ &= \frac{Q^2}{2C} \left(1 + \frac{C_d}{C} \right) + \frac{\Phi^2}{2L} - \frac{C_d}{C} V_s(t) Q + \text{Const} \end{aligned}$$



$$E = \frac{p^2}{2m} + \frac{1}{2} k x^2 - m g x$$

Driven



$$\mathcal{H} = \frac{Q^2}{2C} \left(1 + \frac{C_d}{C} \right) + \frac{\Phi^2}{2L} - \frac{C_d}{C} V_s(t) Q$$

$$= \frac{Q^2}{2C_\Sigma} + \frac{\Phi^2}{2L} - \frac{C_d}{C} V_s(t) Q$$

$$\begin{cases} \hat{a} \sim (\hat{\Phi} + i\hat{Q}) \\ \hat{a}^\dagger \sim (\hat{\Phi} - i\hat{Q}) \end{cases} \leftrightarrow \begin{cases} \Phi \sim (\hat{a}^\dagger + \hat{a}) \\ Q \sim i(\hat{a}^\dagger - \hat{a}) \end{cases}$$

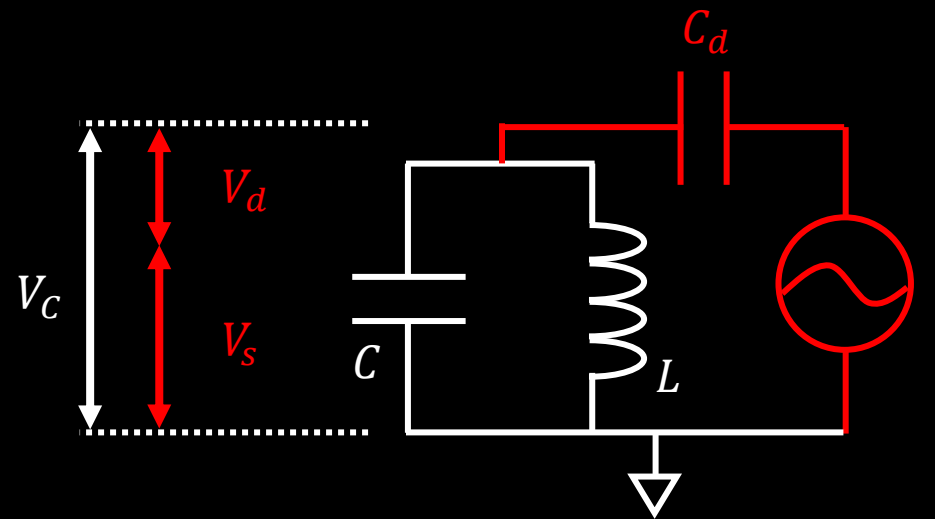
$$= \hbar\omega_\Sigma \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \hbar C_d V_s(t) \sqrt{\frac{\omega_\Sigma}{2\hbar(C+C_d)}} * i(\hat{a}^\dagger - \hat{a})$$

$$V_s(t) = V_0 * f(t)$$

$$= \hbar\omega_\Sigma \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \hbar C_d V_0 \sqrt{\frac{\omega_\Sigma}{2\hbar(C+C_d)}} * f(t) * i(\hat{a}^\dagger - \hat{a})$$

Rabi Frequency Ω

Driven



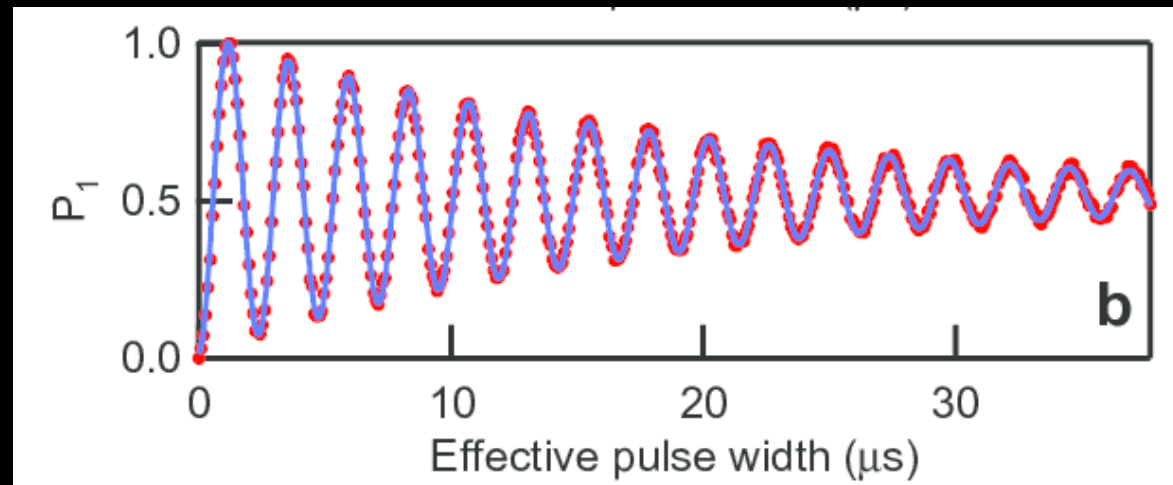
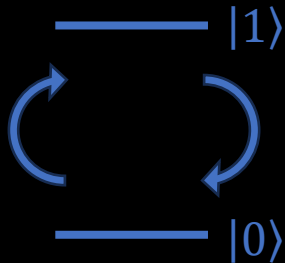
$$\mathcal{H} = \hbar\omega_{\Sigma} \left(\hat{a}^{\dagger} \hat{a} + \frac{1}{2} \right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_{\Sigma}}{2\hbar(C + C_d)}}$$

Informally, for 2-state system,

$$(\hat{a}^{\dagger} - \hat{a})|0\rangle = \hat{a}^{\dagger}|0\rangle = |1\rangle$$

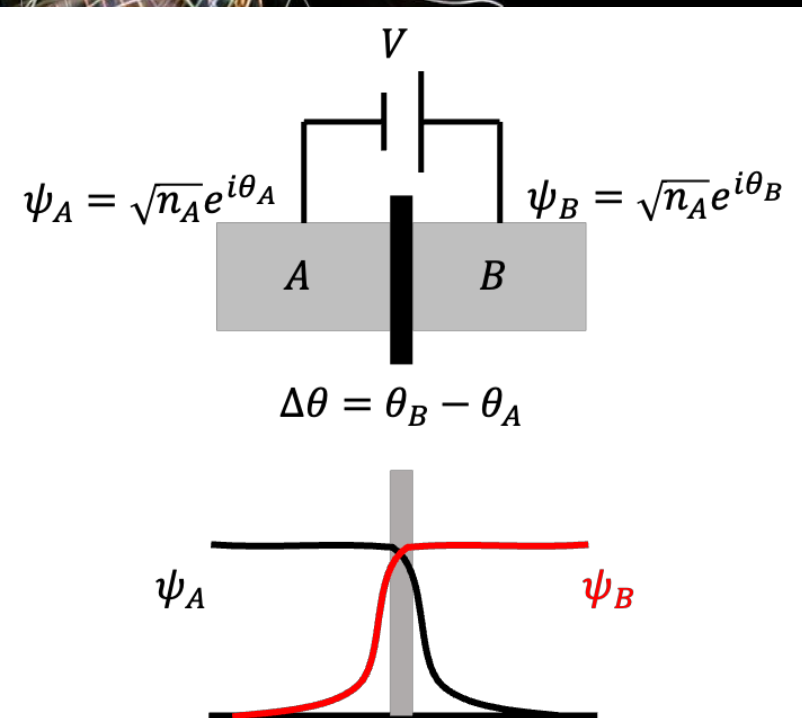
$$(\hat{a}^{\dagger} - \hat{a})|1\rangle = -\hat{a}|1\rangle = -|0\rangle$$





超導量子位元 Transmon

超導量子位元-非線性LC電路



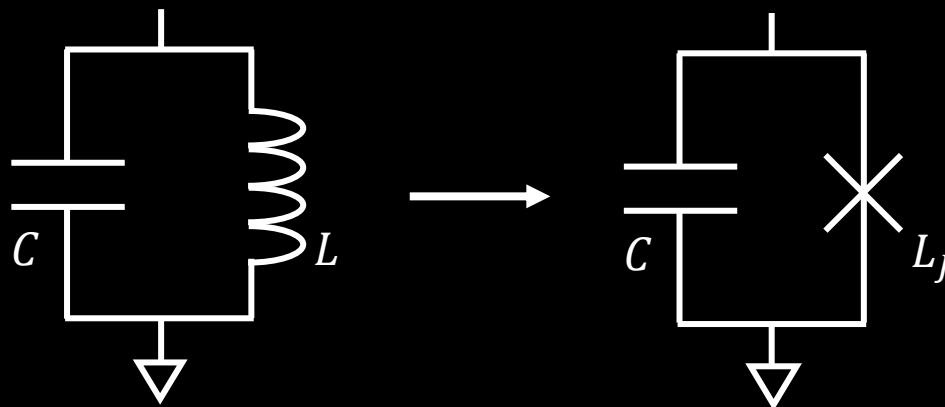
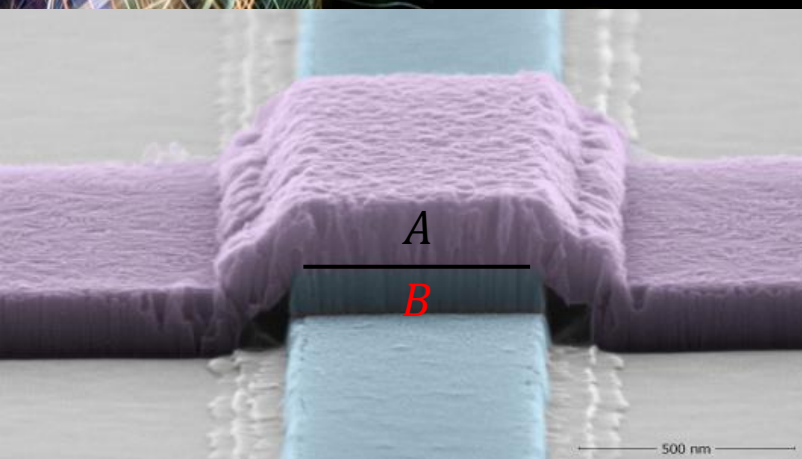
$$\Delta\dot{\theta} = \frac{2e}{\hbar} V$$

$$I = I_c \sin(\Delta\theta)$$

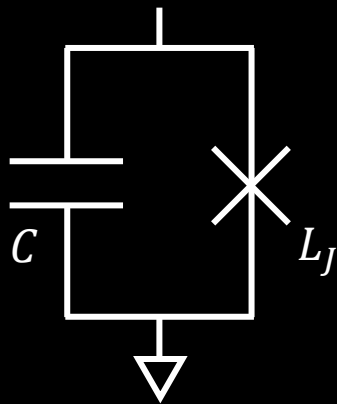
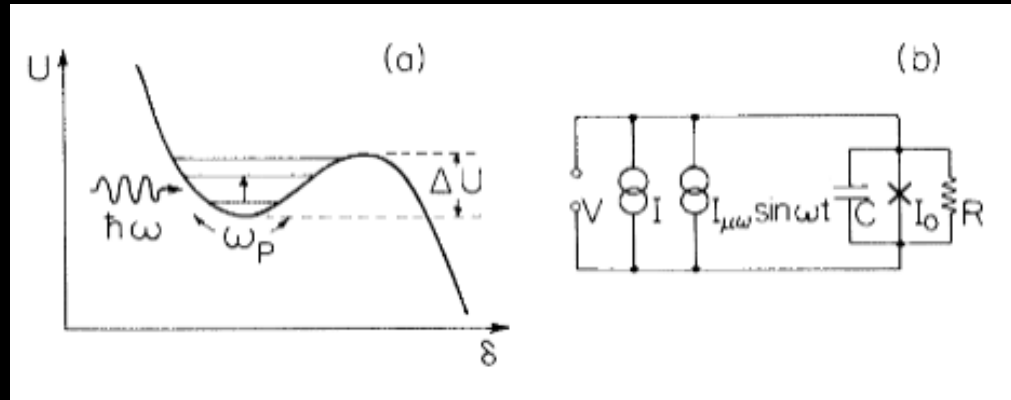
$$\dot{I} = I_c \cos(\Delta\theta) * \Delta\dot{\theta} = I_c \cos(\Delta\theta) * \frac{2e}{\hbar} V$$

$$V = \frac{\hbar}{2e} \frac{1}{I_c \cos(\Delta\theta)} \dot{I} = \frac{h}{2e} \frac{1}{\sqrt{I_c^2 - I^2}} \dot{I} \equiv L_J(I) \dot{I}$$

$$\omega = \frac{1}{\sqrt{L_J C}} \propto \sqrt[4]{I_c^2 - I^2}$$



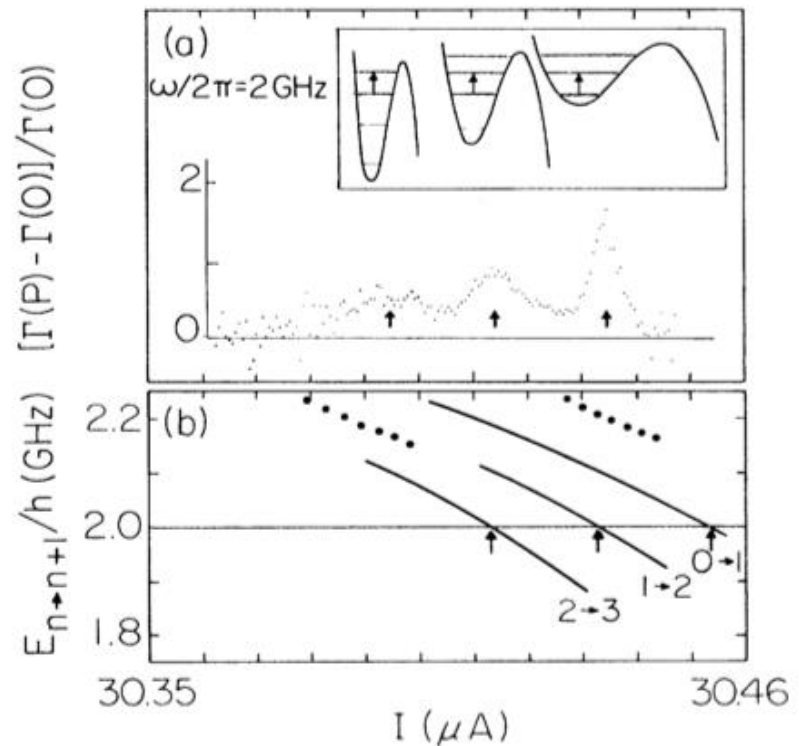
超導量子位元-非線性LC電路



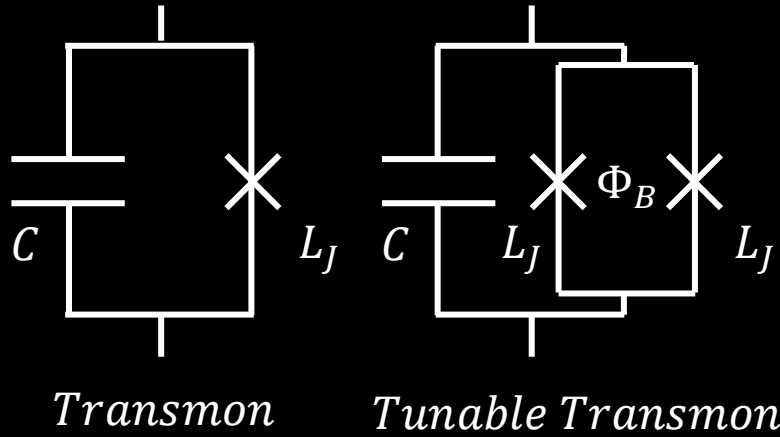
1985

$$\omega = \frac{1}{\sqrt{L_J C}} \propto \sqrt[4]{I_c^2 - I^2}$$

10.1103/PhysRevLett.55.1543, Nobel



Josephson Junction Loop



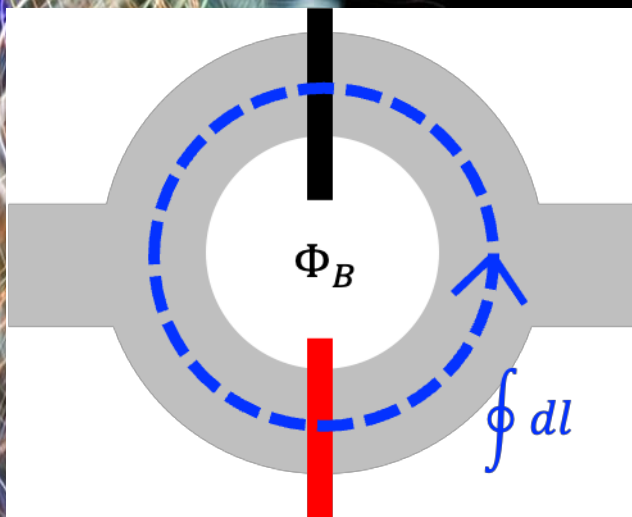
$$\Delta\dot{\theta} = \frac{2e}{\hbar} V$$

$$I = I_c \sin(\Delta\theta)$$

$$\dot{I} = I_c \cos(\Delta\theta) * \Delta\dot{\theta} = I_c \cos(\Delta\theta) * \frac{2e}{\hbar} V$$

$$V = \frac{\hbar}{2e} \frac{1}{I_c \cos(\Delta\theta)} \dot{I} = \frac{h}{2e} \frac{1}{\sqrt{I_c^2 - I^2}} \dot{I} \equiv L_J(I) \dot{I}$$

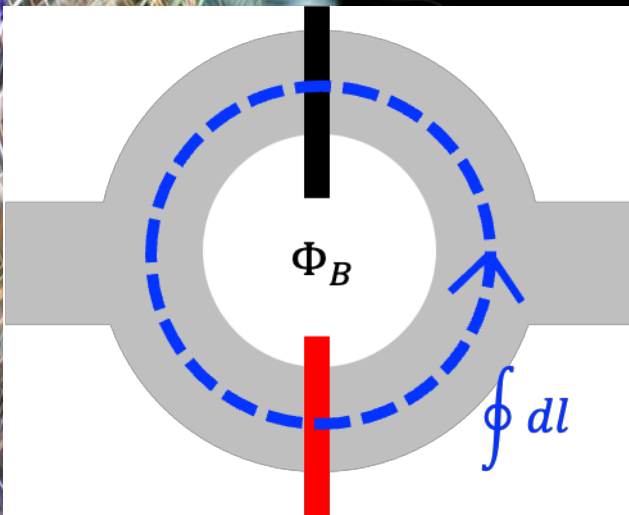
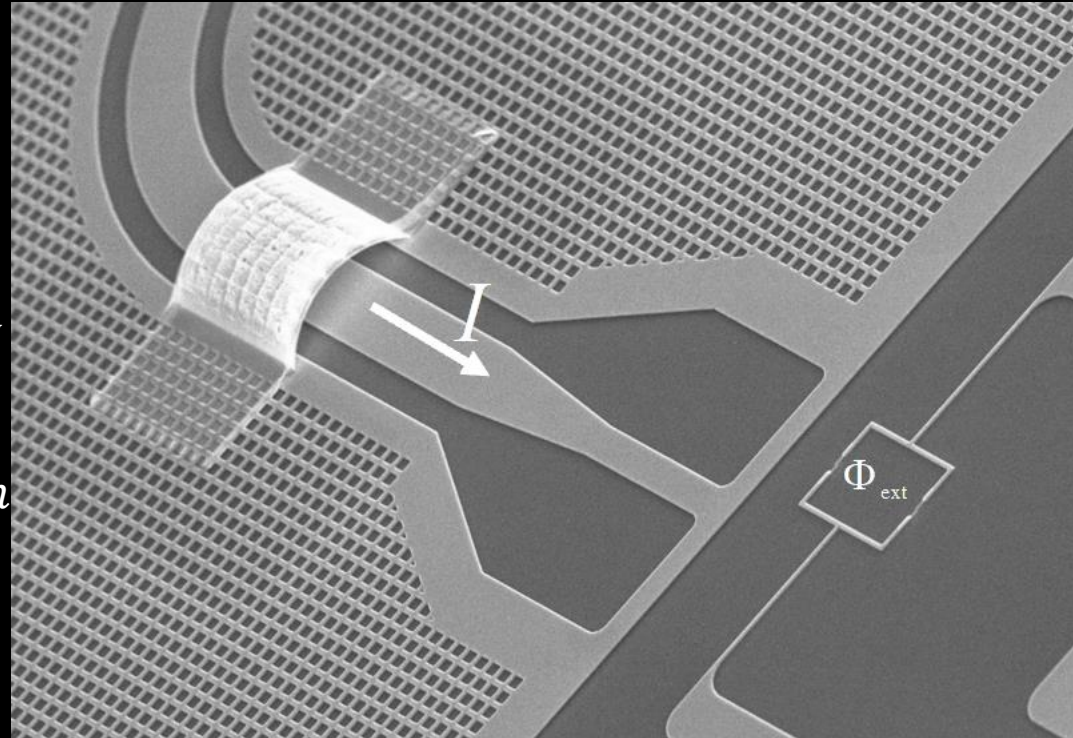
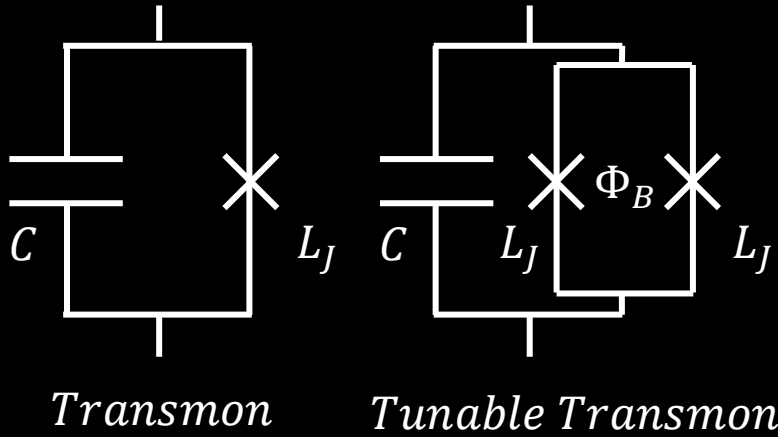
$$\omega = \frac{1}{\sqrt{L_J C}} \propto \sqrt[4]{I_c^2 - I^2} \propto \sqrt{\cos(\Delta\theta)}$$



$$\omega \propto \sqrt{\cos\left(\Delta\theta + \frac{\Phi_B}{\Phi_0}\right)}$$

$$\Phi_0 = \frac{2\pi\hbar}{2e}$$

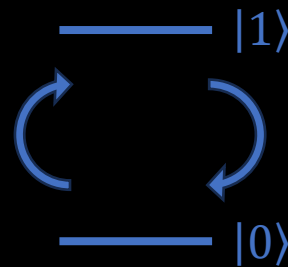
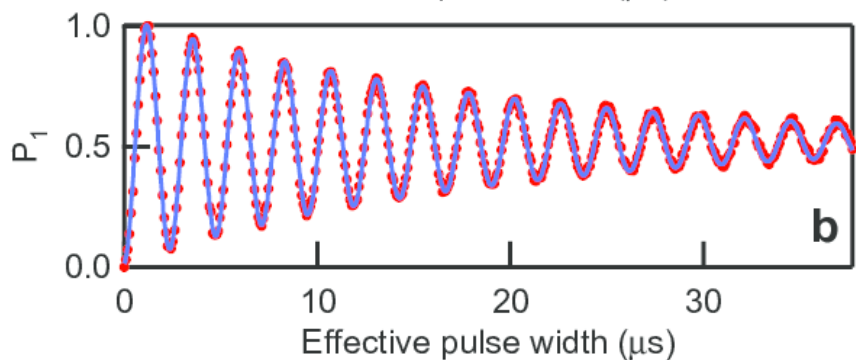
Josephson Junction Loop



$$\omega \propto \sqrt{\cos\left(\Delta\theta + \frac{\Phi_B}{\Phi_0}\right)}$$

$$\Phi_0 = \frac{2\pi\hbar}{2e}$$

Chevron



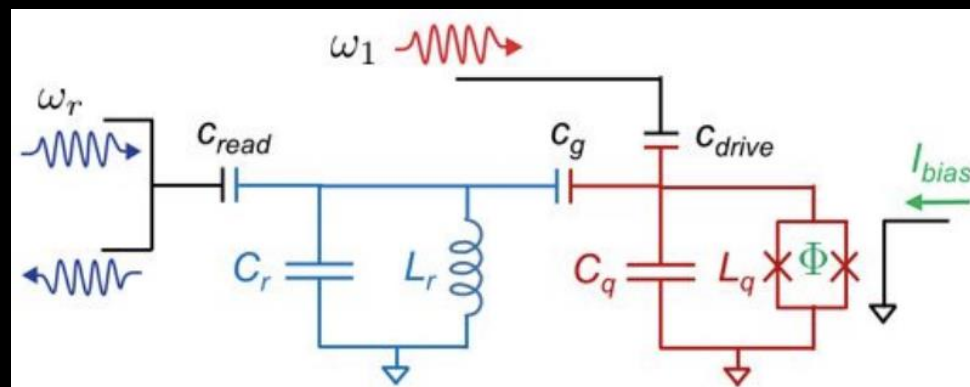
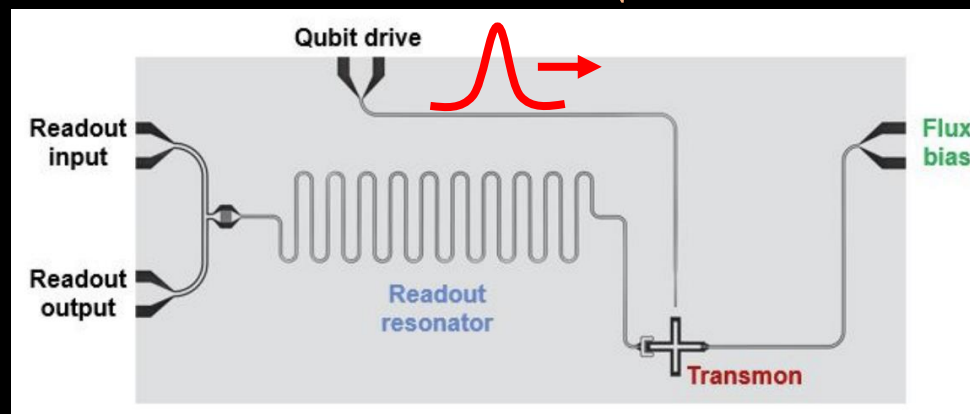
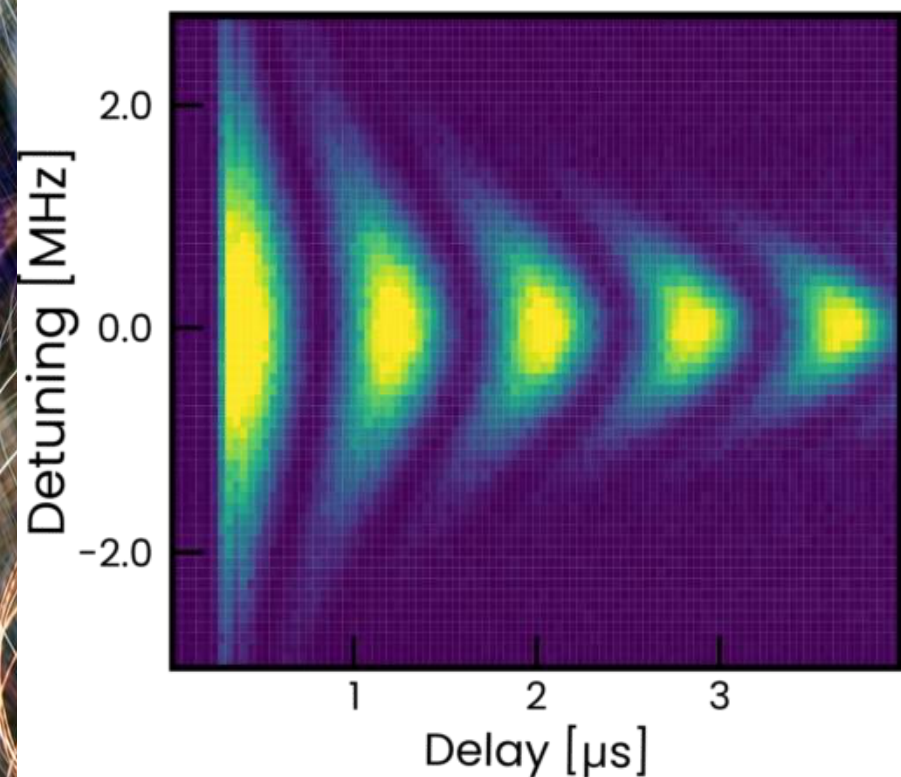
$$\omega_{\Sigma} \propto \sqrt{\cos\left(\Delta\theta + \frac{\Phi_B}{\Phi_0}\right)}$$

$$\Phi_0 = \frac{2\pi\hbar}{2e}$$

$$\mathcal{H} = \hbar\omega_{\Sigma}\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right) + \hbar\Omega * f(t) * i(\hat{a}^{\dagger} - \hat{a})$$

$$\text{Rabi Frequency } \Omega = C_d V_0 \sqrt{\frac{\omega_{\Sigma}}{2\hbar(C + C_d)}}$$

RABI-CHEVRON





Thank you!